

HUNTER POWELL

hunterpowell99@gmail.com • github.com/hunterpowell • hunterpowell.dev • 801-712-0183

EDUCATION

California State University, Sacramento - B.S. Computer Science (Expected December 2026) GPA: 3.5

Relevant Courses: Machine Learning, Parallel Computing, Data Structures, Algorithms, Networking, OS

TECHNICAL SKILLS

- **Languages:** Python, C++, CUDA, Java, JavaScript, Rust, Typescript
- **Frameworks/Tools:** Django, OpenMP, CMake, Numpy, Pandas, scikit-learn, Git, Axum, Tokio
- **Databases/Other:** MySQL, REST APIs, Stripe integration

PROFESSIONAL EXPERIENCE

Software Engineer — Lydia's Law Site | Sacramento, CA 2025-2026

- Full-stack Django app built from scratch by an 8-person team over 8 months for a family law practice.
- Collaborated on the 5-app modular architecture and 3-tier role-based access control (Guest / Client / Admin), contributing to shared infrastructure including Stripe/Calendly webhook handling and Google OAuth authentication.
- Achieved 90% code coverage using Django's built in testing framework.
- Migrated site content from hardcoded templates into a normalized MySQL schema, refactoring Django templates to render dynamically from the database.

PROJECTS

Genetic Algorithm - C++, OpenMP, Python github.com/hunterpowell/GeneticAlg

- Simulated the "coverage problem" - given a robot with limited energy, evolve a set of movement rules that maximizes how much of an unknown grid it can explore before running out of power.
- Encoded each robot's behavior as 32 conditional rules (genes) mapping perceived neighbor states → movement direction; robots sense N/E/S/W cell states and fire the first matching rule each turn.
- Applied elitism, tournament selection, uniform crossover, and random mutation across 500-agent generations, causing robots to independently converge on a boustrophedon (row-sweep) pattern achieving >90% average map coverage.
- Parallelized evaluation and repopulation loops with OpenMP, achieving a 27x speedup (138s → 5s) for 500-agent, 500-generation simulations.

Slate Virtual Tabletop – Rust, Axum, Tokio, Typescript github.com/hunterpowell/SlateVTT

- Real-time multiplayer virtual tabletop for remote D&D — shared map, token movement with per-owner permissions, initiative tracking, and fog of war — replacing Foundry VTT for an active 6-player group; self-hosted with no database, room state persisted as a JSON snapshot and restored on restart.
- Architected the server as a single-owner room actor on Tokio: all mutations serialize through one task owning RoomState, and every outbound WebSocket message is filtered per recipient, so hidden monsters, monster HP, and unexplored map are never sent to a player's client.
- Implemented server-authoritative visibility from DM-traced walls and doors — line of sight computed per player each move, unioned into remembered explored area, with a DM override to reveal or black out regions by hand.

WORK EXPERIENCE

Barista | Starbucks | 2018–2019

- Delivered fast, accurate service to 100+ customers daily while collaborating with a 5-person team to improve efficiency.

Automotive Technician | Havoline Xpress Lube | 2015–2017

- Performed diagnostics and maintenance on 60+ vehicles daily, following safety protocols and communicating clearly with clients.
- Acted as senior technician among peers for the final year, informally mentoring newer hires and serving as the go-to for complex diagnostics.